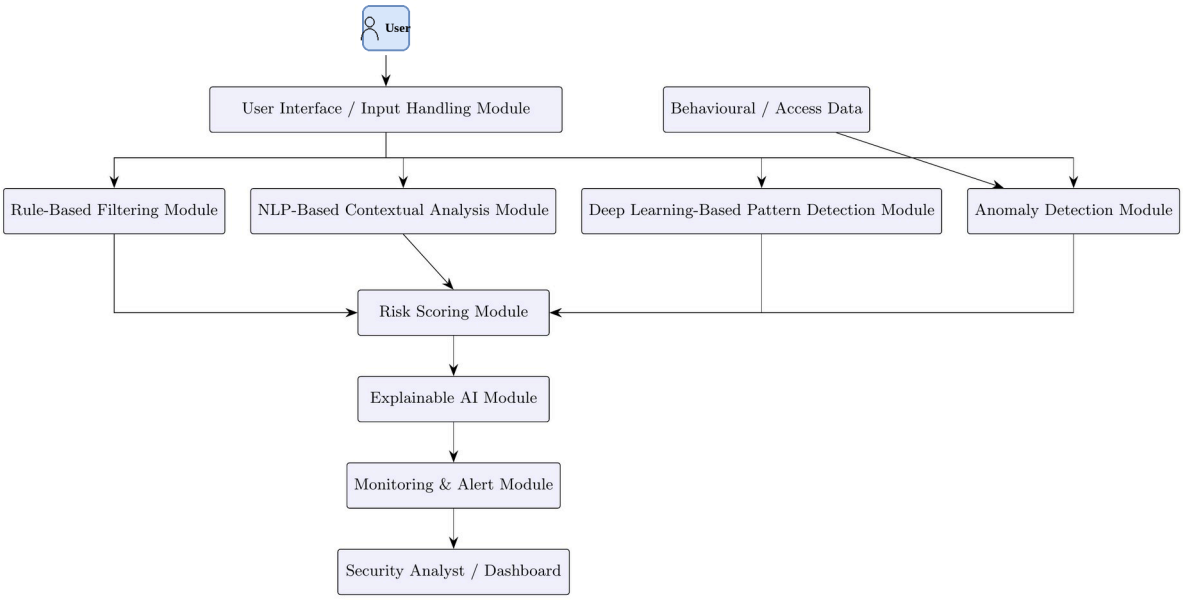

Fig. 5.1: System Architecture of the Intelligent Personal Data Leakage Detection System

5.2 Data Flow Diagram (DFD)

This section illustrates the data flow diagrams of the proposed system, showing the complete movement of data throughout the personal data leakage detection process, from the context-level view down to the internal steps of its more complex processes.

5.2.1 DFD Level 0

At Level 0, the system is represented as a single process. The user or an integrated platform submits text data or a document as input. The system processes this input through its combined rule-based, NLP, deep learning and anomaly detection pipeline and returns a risk score, an alert (where applicable) and an explanation report as output to the security analyst, as shown in Figure 5.2.

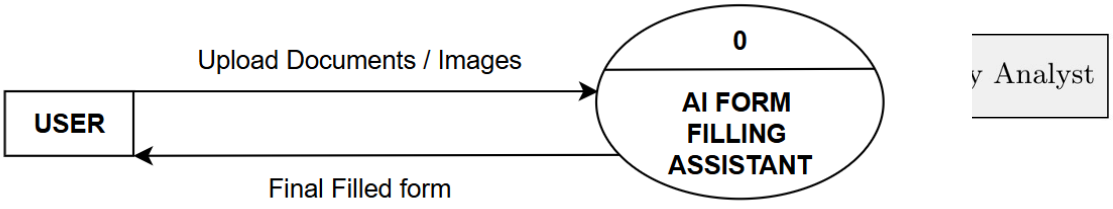

Fig. 5.2: DFD Level 0 of the Intelligent Personal Data Leakage Detection System

5.2.2 DFD Level 1

At Level 1, shown in Figure 5.3, the internal processing of the system is broken down further. The process begins when the input is received and validated by Process 1.0 (Input Validation). The validated input is then passed to Process 2.0 (Rule-Based Filtering), which draws on the calibration dataset (D1) for its configured patterns, and